Owais Ali Khan

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EDUCATION

FAST NUCES, KARACHI

• Bachelors of Science in Computer Science, CGPA: 3.88

ACHIEVEMENTS

- Gold Medallist ICPC Asia Topi Regional '24-25 (National 4th rank)
- Silver Medallist ICPC Asia Topi Regional '23-24 (National 8th rank)
- IEEE Xtreme 17.0 (Global Rank 531/4200)
- Rector's List (Spring '22)
- Dean's List (7 semesters, Fall '21 -- Spring '25)

SKILLS

LANGUAGES

Java • Python • C/C++ • SQL • Go • Bash

BACKEND & ML

Django • FastAPI • REST APIs • TensorFlow/Keras

DATABASES & MESSAGING

PostgreSQL • MySQL • Redis • Elasticsearch • Kafka

DEVOPS & CLOUD

Docker • Kubernetes • Jenkins (CI/CD) • AWS (S3, ECS, EKS, ECR) • NGINX • Git • Linux

SECURITY

CVE remediation (Trivy, Grype) • Container hardening

VOLUNTEERING & COMMUNITIES

- Student Lab Assistant, DSA (Fall '24, Fall '25)
- ACM Devday'25 Mentor (Jan '25 - Apr '25)
- Procom'24 Co-head CS Competitions (Jan '24 - Mar '24)
- IEEE Membership (Dec '23 - Dec '24)

EXPERIENCE

SECURITI AI | Associate Backend Engineer

June '25 - Present | Hybrid

- Developed and maintained backend **Java** and **Python** services detecting *PII/sensitive data* in unstructured files and databases at enterprise scale.
- Engineered *file-format* extraction and **OCR** across PDF, MS Office, OpenDocument and email formats, hardening document *parsers* against malformed and adversarial inputs.
- Built and deployed a new **gunicorn**-based **Python** microservice with multi-stage **Docker** builds, boosting detection-pipeline throughput and scalability.
- Re-architected state storage on **Redis** and hardened **Kubernetes** workloads (non-root, read-only FS, dropped capabilities); fixed **ReDoS** flaws, drove **CVE remediation**, and root-caused production **OOMs** and pod crash-loops under strict *SLAs*.

SECURITI AI | Software Engineer Intern

June '24 - Aug '24 | Hybrid

- Engineered high-performance infrastructure for large-scale *regular-expression* matching using **Intel Hyperscan**, achieving speedups of 2–3 orders of magnitude.

RETROCAUSAL AI | Applied Research Intern

Mar '24 - June '24 | Remote

- Developed *RESTful* API endpoints and automated reporting workflows using the **Django REST Framework**, with scalable storage and retrieval on **Amazon S3**.

LDP LOGISTICS | IT Support Specialist

June '23 - Sep '23 | Onsite

- Built and deployed a full-stack *web application* (**Django**, **Bootstrap4**) for managing electronic *inventory* across local and overseas offices.

PROJECTS

LLM TEXT DETECTOR | Natural Language Processing

Fall 2024, Spring 2025 | Final Year Project

- **A Hybrid Approach to LLM-generated Text Detection**: combined *cross-perplexity* scores with a fine-tuned *T5* transformer to detect AI-generated text.
- Achieved a 94% **F1-score** on a curated dataset (7500 generations per class).

POLARIS | High-performance Image Processing

Fall 2023 | Parallel and Distributed Computing

- A *parallelized* image-enhancement program implementing **Gaussian filtering** and **Histogram Equalization**.

LDP ASSET MANAGER | Fullstack Web Application

Summer 2023 | Full Stack Development

- An inventory-management *web portal* (**Django**, **MySQL**, **MUI**) tracking assets, accessories and licenses across all LDP offices.